نقل الكهرباء
National Grid SA

ORIGINAL DOCUMENT APPROVED: October 16, 2006

REVISIONS

No.	Date	DESCRIPTION
1	Jan. 31, 2-012	Changes in the International Standards, modification of existing clauses and added new clauses
2	May 31, 2018	Technical requirements of LED lamps are added
3		

TES-P-119.25

Revision No. 02

LIGHTING AND RECEPTACLES FOR SUBSTATION YARD & BUILDING

PROPRIETARY INFORMATION

This document contains proprietary information developed by and for exclusive use of National Grid Saudi Arabia. Your acceptance of the document is an acknowledgement that it must be used for the identified purpose/application and during the period indicated. It cannot be used or copied for any other purposes nor released to others without prior written authorization of National Grid Saudi Arabia. National Grid Saudi Arabia will not be responsible for misuse and/or misapplication, and any harm resulting there from. National Grid Saudi Arabia also reserves the right to take any necessary actions to protect its interest against unauthorized use.

ORIGINATED BY	CERTIFIED BY	APPROVED BY
Engr. Khaled S. Al-Mutairi Division Manager, Substations Specifications	**Engr. Suliaman N. Al-Dosari** Department Manager, Standards & Specifications	**Engr. Waleed A. Al-Saadi** Vice President, Engineering
Date of Approval: May 31, 2018		

TRANSMISSION ENGINEERING STANDARD

TABLE OF CONTENTS

1.0 SCOPE

2.0 GENERAL

3.0 YARD LIGHTING

3.1 Design criteria
3.2 Lamps, Fixtures and Ballasts

4.0 BUILDING LIGHTING

4.1 General Lighting
4.2 Exit and Emergency Lighting
4.3 Lamps, Fixtures and Ballasts

5.0 ILLUMINATION LEVELS

6.0 POWER RECEPTACLES

6.1 Outdoor Power Receptacles
6.2 Building (Indoor) Receptacles Outlets

7.0 LIGHTING PANELS

8.0 WIRES, CABLES, CONDUITS AND JUNCTION BOXES

9.0 GROUNDING OF LIGHTING EQUIPMENT

10.0 BIBLIOGRAPHY

1.0 SCOPE

This Engineering Standard specifies general guidelines for design, selection and installation of energy efficient lighting equipment and design of outdoor (substation yard), indoor (building) lighting systems and indoor/outdoor power receptacles intended to be used in the Transmission system of National Grid Saudi Arabia.

2.0 GENERAL

2.1 The lighting installation shall be designed and installed as per the latest editions of:

a. National Electrical Code (NEC): NFPA 70

b. National Electrical Safety Code: ANSI/IEEE C.2

c. IES Lighting Handbook

2.2 Latest revision of National Grid Saudi Arabia Standards, Specifications and Industrial standards shall be referred.

2.3 The lighting installation shall be designed with particular regard to ease of subsequent maintenance (lamp cleaning and replacement) with minimum requirement of special tools or equipment. Consideration shall also be given to minimize the necessity of substation equipment outage.

2.4 All wiring and lighting equipment shall be suitable for working safely at full load at the indoor/outdoor ambient conditions as specified in 01-TMSS-01. Neutral conductors shall be of the same cross sectional area as phase conductors throughout the lighting and small power installations.

2.5 Outdoor yard, perimeter, gate house, check point and Company's signboard lighting shall be fed from 230 V_{ac} supply through MCCBs (Moulded Case Circuit Breakers) in 400/230 V_{ac} (L-L/L-N Nominal), 3 phase, 4 wire AC distribution panel. Yard lighting luminaries shall be connected from the AC distribution panel such that 50% illumination level can be obtained by manually switching off the circuits, whenever required. Multi-conductor cables shall be used for this purpose. The MCCB (TPN) shall have a minimum current rating of 63 A and an interrupting current rating of 20kA. MCCB shall be temperature compensated and de-rated according to the ambient temperature.

2.6 The general indoor lighting, 230 V_{ac} outdoor/indoor receptacles, panel lighting and panel space heaters shall be supplied from 400/230 V_{ac}, 3 phase, 4 wire AC lighting panels.

2.7 Miniature circuit breakers (MCBs) of required voltage and current rating and short circuit current interruption capability, sized to ballast manufacturer's recommended full load current-carrying-capability shall be provided for each circuit. The MCBs shall have thermal-magnetic overload protection and ON/OFF status indicator. The MCBs shall be temperature compensated and de-rated for the ambient temperature.

2.8 MCBs and MCCBs shall be as per 37-TMSS-03.

3.0 YARD LIGHTING

3.1 Design criteria

3.1.1 The yard (outdoor) lighting shall be designed by surface discernment method to:

a. permit rapid security surveillance inspections

b. make possible thorough equipment status inspection

c. ensure the safe performance of normal and emergency switching

3.1.2 Beam-lumen method shall be used to estimate the quantity of flood lights needed; but to check for uniformity, point-method shall be used.

3.1.3 For the purpose of lamp lumen calculations, consider the following typical values of light-loss-factors viz. Lamp Lumen Depreciation (LLD) and Luminaire Dirt Depreciation (LDD), unless available from lamp and fixture manufacturer:

Table 25-1: Light Loss Factors

Type of Lamp	LLD	LDD
High Pressure Sodium (HPS) Lamp Wattage - 250 to 1000W	0.87	0.71
LED Lamps- 100 to 250W	0.70	0.71

3.1.4 The choice of mounting height shall be determined by the area to be covered. The mounting height for yard lighting shall not be less than 10 meters. Flood lights shall be mounted on painted steel poles. Additional flood lights may be mounted on brackets from the different sides of the steel transmission tower/gantry structures, on the building top, wherever necessary. Mounting of flood lights on steel transmission tower/gantry shall be avoided as much as possible to facilitate better maintenance accessibility to the fittings. Proper working clearance from live parts and the fittings shall be maintained. Localized bracket/wall mounted flood lights shall be provided around transformers, etc.

3.1.5 The lighting poles shall be selected based on weight of all luminaries, brackets, other accessories, etc and effective projected area (EPA), i.e. actual projected area of the luminaire multiplied by the coefficient of drag.

3.1.6 The location of lighting pole shall not interfere with duct banks, overhead lines, substation equipment and movement of vehicles and heavy equipment.

3.1.7 If there is a particular part of the outdoor yard area, away from the pole, that needs to receive the highest possible level of illumination, the flood light mounting height shall be so selected that the flood light can be aimed close to an angle of 54.7 degrees to Nadir (Vertical).

3.1.8 Each lighting/lightning steel mono pole shall have an accessible compartment lockable with keys, MCBs or fuses, necessary terminal blocks, gland plate and other control equipment, as required. The poles shall be fully protected and sealed against ingress of water or vermin.

3.1.9 Control of yard lighting circuits shall be accomplished with electrical contactors and photo cells (with auto/manual changeover facility for manual control), connected to a continuously energized circuit. Also, continuously energized outdoor convenience outlets/receptacles for supplemental portable maintenance or repair-work lighting shall be furnished on the lighting poles without requiring separate circuits as per drawing # TA-800004.

3.1.10 Photo cells shall be mounted at a high level in a position free from shadows and not facing any lighting fixtures directly, but accessible for maintenance. The “auto/manual” change-over switch shall be provided at 1.2 meters above finished floor level.

3.1.11 The security perimeter, gatehouse, check points lighting in the substation shall be as per SEC-04 of High Commission for Industrial Security (HCIS).

3.1.12 DC emergency lights need not be provided for yard lighting.

3.2 Lamps, Fixtures and Ballasts

3.2.1 Where operations do not call for color-critical-judgment, color-improved 250W/400W/1000W, high pressure sodium vapour lamp (HPS) or equivalents, such as LED street light fitting shall be used for yard lighting. The minimum lamp efficacy shall be 110 lumens/watt for HPS lamp and 100 lumens/watt for LED lamp including ballast losses. Fixture certificate indicating performance standard and applicable IEC standard or equivalent shall be submitted. Applicable IEC or equivalent standard shall be permanently labeled on the lighting equipment.

3.2.2 COMPANY’s sign board lighting shall be provided by two (2) 250 W flood lights or LED lights as per clause 3.2.4, controlled by photo cell.

3.2.3 The high pressure sodium vapour lamp and fixture shall have the following characteristics:

a. Tubular double-ended clear polycrystalline alumina arc tube housed in translucent borosilicate glass outer bulb.

b. Initial lumen of not less than 27,500 lumens for 250 W clear HPS, 50,000 lumens for 400 W clear HPS and 140,000 lumens for 1000 W clear HPS lamps.

c. Fixture shall be one piece, IP54 (per IEC 60529) enclosed/gasketted extruded aluminum body with an asymmetrical faceted ALGLAS finish reflector profile and tempered dished clear cover glass.

d. Control gear box shall be die-cast copper free aluminum alloy with hinged/removable door frame and integral with the luminaire. Door fastening device shall be captive.

e. Rated life of the HPS lamp based on 10 hours per start shall be minimum 24000 hours.

f. HPS lamps shall have mogul base and minimum stroboscopic effect. The lamp output wattage shall not vary more than ±10% of the normal wattage.

3.2.4 LED street light shall have the following characteristics:

a. LED Street and yard light shall be with integral drive ballasted LED module constant current type suitable for $230V_{ac}$, 60Hz, power factor (≥ 0.90) in ambient conditions (-) 5 ℃ to (+)55 ℃ for 11 hours per day, humidity ≥ 90% with Luminous flux output not less than 20,000 lm and minimum efficacy 100 lm/watt.

b. Housing shall be metal with powder coated and rust resistant and cover shall be UV stabilized Polycarbonate.

c. The housing design shall allow to provide excellent thermal management to get long LED system life.

d. Cooling shall be heat sink without fan.

e. Degree of protection of drive and Assembly shall be minimum IP 65 and mechanical impact shall be IK 08.

f. LED light shall be with correlated color temperature (CCT) greater than 5000K (cool white).

g. Color rendering index (CRI) shall be ≥ 70.

h. Measured efficacy of LED module shall not be less than 90% of the rated LED module efficacy as declared by the manufacturer.

i. Lumen Depreciation of LED light source shall be minimum 70% of initial lumen for minimum 50,000hrs.

j. LED light Optic shall provide high quality uniform light distribution and better light trespass control.

k. The driver shall be protected against surge voltage, high voltage cut-off, short circuit and temperature.

l. Noise level of Power assembly shall be minimum Class A sound rating per ANSI Standard C63.4 and Electromagnetic compatibility shall be Criterion A as per IEC 61000-4-2.

m. The lamp, fixtures and drive (ballasts) shall be in compliance to respective IEC or equivalent standards.

3.2.5 Flood light fixtures with an asymmetric beam pattern shall be used for better area coverage. The flood lighting fixtures shall be semi cut-off type to control the spill of light and direct glare properly. They shall have beam spread of NEMA 6 × 5 as per NEMA FA1-1973 or equivalent.

3.2.6 The fixtures shall have corrosion-resistant external hardware, heavy duty stainless steel trunnion with a degree indicator and reposition stop, an activated charcoal adsorptive filtered vent and built-in aiming sight and provision for fitting mounting adapter. The door assembly shall include mechanical stops to assure proper compression of the hollow core silicone/neoprene gasket. The fixtures shall be painted as per 01-TMSS-01.

3.2.7 The ballast/driver shall be 230 V_{ac}, constant wattage auto (CWA) regulator of compensated high power factor (HPF) type having input voltage range of ±5%, a minimum lamp-ballast combination P.F. of 0.90 and with crest factor of lamp current shall not higher than 1.8. The ballast shall have current total harmonic distortion (THD) below 20%.

3.2.8 The ballast/driver (CWA) must be capable of riding out expected dips in the line voltage without extinguishing lamps. The insulation system of the core-and-coil ballast shall be designed for a maximum 180ºC hot-spot temperature. The temperature rise of the ballast coils plus the maximum ambient temperature must not be greater than 165ºC.

3.2.9 There shall not be any objectionable radio interference from the ballast during normal operation.

3.2.10 The ballast/driver shall be rated for the outdoor ambient temperature and shall have low power loss.

3.2.11 The lamps, fixtures and ballasts shall be as per relevant IEC or equivalent standards.

4.0 BUILDING LIGHTING

4.1 General Lighting

4.1.1 The building (indoor) lighting shall be complete with lighting fixtures, switches, receptacles, lighting panels, conduits, necessary cabling and other accessories.

4.1.2 The following formula per lumen method shall be used for general lighting to estimate the quantity of fluorescent/LED lamp fixture (N) needed to illuminate the indoor areas within the substation building:

$$N = \frac{E_{maintained} \times \text{Workplane area}}{\text{Lamps per luminaire} \times LL \times CU \times LLF} \qquad \text{(Eq. 25-1)}$$

where,
- N = Number of luminaries or fixture
- E = Illumination
- LL = Initial Rated Lamp Lumens
- CU = Coefficient of Utilization
- LLF = Light-Loss Factor (i.e. LLD x LDD x ballast factor)
- LLD = Lamp Lumen Depreciation
- LDD = Luminaire Dirt Depreciation

4.1.3 The zonal cavity method as per IES lighting handbook shall be used for determining the coefficient of utilization (CU) in indoor lighting calculations. If the ceiling, floor or room area is broken up with beams, ducts, machinery or other obstructions, then the area to be lighted should be broken down into smaller sections for design purpose. To read the manufacturer's CU table, the room cavity ratio, wall reflectance, effective ceiling and floor cavity reflectance must be known as well as the tentative lamp and luminaire choice.

4.1.4 If the reflectance are unknown, reflectance of 30% for wall, 30% for ceiling and 20% for floor shall be assumed.

4.1.5 For illuminating office areas, Equivalent Sphere Illumination (ESI) concept can be used, wherever necessary. To prevent shadows, luminaries spacing shall be close to the mounting height. Recommended surface reflectance for offices per IES lighting handbook shall be considered in designing office lighting.

4.1.6 The IES recommended maximum luminance ratio as per American National Standard Practice for Industrial Lighting, A11.1-1973, RP-7, shall be followed for all workplace lighting within the substation building.

4.1.7 Level of illumination and uniformity in lighting is to be checked by point-method as per IES lighting handbook, utilizing the inverse square and cosine law.

4.1.8 Illumination shall be as uniform as possible and free from obtrusive glare. Illumination levels generally shall be measured in a horizontal plane at 1 (one) meter above the floor level. Illumination levels at control and instrument panels shall be measured on the vertical plane at the panel position.

4.1.9 Special care shall be taken in selection of fittings and diffusers within the control room and other rooms within the building and also about

illumination of all control, instrument and equipment panels to avoid glare and reflections from instrument glasses.

4.1.10 UGR (Unified Glare Rating) method shall be used for calculating glare from luminaries and calculated glare index shall not exceed limits as per EN 12464-1.

4.1.11 The mounting height (MH) of the fixtures over batteries, work benches and desks shall be 2440mm, elsewhere 2750mm, above the finished floor level.

4.1.12 Proper fixture horizontal spacing "(S)" versus mounting height "(MH)" ratio shall be selected from the manufacturer's catalog to get the best utilization and uniformity in lighting.

4.1.13 The location of lighting fixtures shall not interfere with cable tray or HVAC ducts or fire alarm system. A superimposed drawing with all the system shall be submitted to reflect the proper locations in order to avoid any obstructions.

4.1.14 Building (indoor) lights including lights in the cable cellar and rooms with more than one entrance shall be controlled by suitable wall mounted "3 way or 4 way switch" adjacent to the main and other entrance doors (when Auto controller is used to control the lights, the switches shall selected according to the requirements). Unless otherwise specified, the wall mounted lighting switches shall be at 1.2 meters above finished floor level. The switch boards shall have cover (insulated) plates and shall be complete with metal boxes, etc. Lighting circuits shall be designed such that no individual light switch is loaded to more than 50% of its nominal rating. Number of lights in one circuit shall be limited such that total darkness shall not occur due to failure of lamps in one circuit.

4.1.15 Wherever switches are specified to be waterproof, they shall be fitted with galvanized covers and mounted in galvanized boxes with gaskets as necessary.

4.1.16 For efficient control of indoor lights in substation logic controller (Auto controller) shall be used with prior approval of scheme by National Grid Saudi Arabia, if requested in TS/SOW. Following minimum requirements shall be provided:

a) This controller is only for AC lights inside the S/S, except the lights in corridors/exit doors/exit access/stairs/aisles.

b) Duration of time out (switch-off) shall be re-settable from "30 minutes to 24 hours" as required.

c) By pass switch shall be provided to by-pass the Controller in emergency cases such as to maintain the light for long time than the pre-set time.

4.2 Exit and Emergency Lighting

4.2.1 Exit light with self-contained sealed lead acid rechargeable standby batteries suitable for three (3) hours illumination, 8 W fluorescent tube or equivalent and charger circuit, connected to a dedicated $230V_{ac}$ branch circuit shall be provided above all the exit doors and at strategic exit locations like that from basement etc. in the substation for safe egress. Explosion proof exit light fittings shall be used in battery room. Exit signs shall conform to section 7.10 of NFPA 101, section 3314 of Uniform Building Code. Exit lighting shall be of single face or double face type. Exit signs lamps shall be provided with direction arrows wherever the direction of travel is not apparent. The batteries shall not be overcharged and shall be protected against short-circuits. The exit lighting shall have push button test switch and AC charge indicator. The exit lighting fixture cover shall be made of plastic with the words "EXIT" and its Arabic equivalent (مخرج). The standard color of the background shall be WHITE and the panel GREEN with WHITE letters. The panel shall be clearly visible under all lighting conditions as per the requirement of clause 6.3.2.4 of 5 star element # 2.24. The height of lettering shall be 152 mm.

4.2.2 DC Emergency lights shall:

a. be fed from DC emergency lighting panel.

b. be provided in each room of the substation building including basement, near exit doors of the substation building and at exit access that includes designated stairs, aisles, corridors and passageways leading to an exit with a facility for manual switching in each room. These MCBs shall be wall mounted outside the room, easily accessible and distinctively identified.

4.2.3 Emergency lighting main supply must be automatically switched on in the event of normal AC lighting failure in the substation per scheme/figure as shown in TES-P-119.30.

4.3 Lamps, Fixtures and Ballasts

4.3.1 35 W, 230 V_{ac}, T5 (high efficacy) fluorescent lamp with G5 bi-pin caps or equivalent, such as LED (technical requirements of LED tubes and lamps are described under clause 4.3.7) and having efficacy of 90 lumens/watt shall be used for general lighting throughout the substation building, unless otherwise specified. The lamps for office/workshop and control room shall have a color rendering index of 85. The fluorescent lamps shall have minimum stroboscopic effect and preferably be instant start type. The T5 lamps shall contain no luminescent material with beryllium, arsenic or cadmium.

4.3.2 60 W, 125 V_{dc} pearl, gas-filled incandescent lamp or LED lamp (technical requirements of LED tubes and lamps are described under clause 4.3.7) shall be used for DC emergency lighting as mentioned in clause 4.2.2. The lamp shall have efficacy of more than 13 lumens/watt.

4.3.3 100 W, 230 V_{ac}, color improved, clear high pressure sodium (HPS) vapour/LED (technical requirements of LED lamps are described under clause 4.3.8) weather proof type lamp or equivalent lamp fitting approved by National Grid Saudi Arabia shall be used for the building exterior lighting over the main entrance of the substation, Substation gate posts, outdoor Transformer Bay (Room), Station Service Transformer Bay, Grounding transformer, Oil immersed reactor, Capacitor bank, DRPC (SVC) and HVDC. The lamp shall be photo electric controlled, automatic with a facility for manual control. The lamp shall have at least 9,500 initial lumens and efficacy of 95 lumens/watt.

4.3.4 Additional 100 W, 230 $V_{ac,}$ color improved, clear high pressure sodium (HPS) vapour /LED (technical requirements of LED lamps are described under clause 4.3.8) high-bay type lamp and fixture may be used for task lighting during maintenance of GIS module wherever necessary. The lamp shall be hanging from the bottom of the crane. The lamp shall have at least 9,500 initial lumens and efficacy of 95 lumens/watt. The HPS fixtures shall have die-cast aluminum housing with acrylic reflectors, heat resistant prismatic glass diffusers and integral control gears. The ballast shall generally conform to clause 3.2.7 & 3.2.8.

4.3.5 All other exterior entrance doors to the substation building shall have external lights having 60 W, 230 V_{ac}, incandescent lamps, mounted over the doors. The lamp shall have minimum efficacy of 13 lumens/watt. LED lamp with minimum efficacy 85 lm/watt shall be used in place of incandescent lamp, as required.

4.3.6 69 kV indoor switchgear cells shall have compact fluorescent fixture having 40 W, 230 V_{ac}, lamp with color rendering index of 85 and electronic ballast, suitably mounted on wall-bracket inside the doors. The lamp shall have minimum efficacy of 87 lumens/watt. LED lamp with minimum efficacy 85 lm/watt shall be used in place of compact fluorescent fixture, as required.

4.3.7 LED tubes and fixture shall have the following characteristics:

a. LED tubes shall be with integral drive ballasted LED module constant current type for use on DC supplies $125V_{dc}$ or AC supplies $230V_{ac}$, 60Hz, power factor (≥ 0.90) in ambient conditions (-) 5℃ to (+)40℃, humidity ≥ 90%, with Luminous flux output not less than “2000 lm and minimum efficacy 85 lm/watt”.

b. Storage temperature shall be minimum (-) 5℃ to (+)60℃.

c. The housing design shall allow to provide excellent thermal management to get long LED system life.

d. Degree of protection shall be minimum IP 20 for indoor.

e. LED light shall be with correlated color temperature (CCT) greater than 5000K (cool white).

f. Color rendering index (CRI-(Ra)) shall be ≥ 80.

g. Measured efficacy of LED module shall not be less than 90% of the rated LED module efficacy as declared by the manufacturer.

h. Lumen Depreciation of LED light source shall be minimum 70% of initial lumen for minimum 50,000hrs.

i. The driver shall be protected against surge voltage, high voltage cut-off, short circuit and temperature.

j. Lamp holder shall be G13.

k. The connection of lamp pins of caps shall not have polarity requirement.

l. Noise level of Power assembly shall be minimum "Class A" sound rating per ANSI Standard C63.4 and Electromagnetic compatibility shall be "Criterion A as per IEC 61000-4-2".

m. Electric shock rating shall be "Class III or above".

n. Lamp shall be without optical risk.

o. The lamp and drive shall be in compliance to respective IEC Standards.

4.3.8 High Bay LED lamps and fixtures for indoor and outdoor application

a. High Bay LED lamps and fixtures shall be with integral drive ballasted LED module constant current type for use on a.c $230V_{ac}$, 60Hz, power factor (≥ 0.90), ambient conditions (-) 5℃ to (+)40℃ for indoor and (-) 5℃ to (+)55℃ for outdoor, humidity ≥ 90%, with Luminous flux output not less than "20,000 lm and minimum efficacy 125 lm/watt".

b. Storage temperature shall be minimum (-) 5℃ to (+)60℃.

c. The housing design shall allow to provide excellent thermal management to get long LED system life.

d. Degree of protection shall be minimum IP 20 for indoor and IP65 for outdoor.

e. LED lamp shall be with correlated color temperature (CCT) greater than 5000K (cool white).

f. Color rendering index (CRI) shall be ≥ 70.

g. Measured efficacy of LED module shall not be less than 90% of the rated LED module efficacy as declared by the manufacturer.

h. Lumen Depreciation of LED light source shall be minimum 70% of initial lumen for minimum 50,000hrs.

i. The driver shall be protected against surge voltage, high voltage cut-off, short circuit and temperature.

j. Noise level of Power assembly shall be minimum "Class A" sound rating per ANSI Standard C63.4 and Electromagnetic compatibility shall be "Criterion A as per IEC 61000-4-2".

k. Electric shock rating shall be "Class III or above".

l. Lamp shall be without optical risk.

m. The lamp and drive shall be in compliance to respective IEC Standards.

4.3.9 Luminaire selection shall be based on ambient conditions, mounting height, beam spread, shielding angle and reflector shape and standby or emergency needs.

4.3.10 The general fluorescent/LED lighting fixtures in the substation building shall be wrap around, wall bracket/ceiling/pendant mounted luminaire with acrylic symmetrical reflector and IP20 (per IEC 60529) design and suitable for 2×35W fluorescent tube lamps/LED lamps (equivalent watt). Luminaire body shall be of sheet steel for general area. General lighting luminaries shall be fitted with clear or opal plastic diffuser with prismatic pattern in bottom and fluted sides. The fluorescent/LED fixtures shall be fully recessed type for rooms having suspended ceiling. In GIS building fluorescent/LED lighting fixtures shall be wall-bracket mounted.

4.3.11 The lighting fixtures and fittings (fluorescent lamp/LED lamp /any other) in the battery room shall be acid vapor-corrosion proof and explosion-proof type. LED fixtures shall be in compliance to clause 4.3.7, with Degree of Protection IP66 and Operating temperature (-) 5℃ to (+)55℃. Light switches / wall mounted push button test switch for battery room shall be installed just outside the battery room. The luminaries in battery room shall have clear polycarbonate cover and the luminaire body shall be of glass reinforced polyester. No receptacle shall be provided inside the battery room.

4.3.12 The DC incandescent lamp fixtures/LED fixtures shall be wall bracket/pendant/ceiling mounted of IP21 (per IEC 60529) design and complete with opal diffusers and suitable reflectors.

4.3.13 The HPS fixtures/LED lamps for building exterior lighting over the main entrance shall be IP54 minimum (per IEC 60529), weather proof type, wall mounted, having die-cast aluminum housing with acrylic reflectors, heat resistant prismatic glass diffusers and integral control gears. The ballast shall generally conform to clause 3.2.7 & 3.2.8.

4.3.14 The AC incandescent lamp fixtures/LED light fixtures for other exterior entrance doors shall be IP54 minimum (per IEC 60529), weather proof type, wall mounted, complete with thermopal glass and die cast aluminum housing and canopy.

4.3.15 The automatic exit light fixture shall have welded steel housing with acrylic diffuser and shall have electronic ballast and integrated circuit charging and IP21 (per IEC 60529) design.

4.3.16 The ballast for fluorescent tube lighting/LED lights shall be of integral 230 V_{ac}, electronic ballast having current total harmonic distortion (THD) not more than 10%. The electronic ballast shall have a power factor of 0.90. The ballast shall not generate objectionable radio frequency interference.

4.3.17 The lighting system corrected noise criterion (LS-NC) rating of the luminaire (i.e. lamp/ballast assembly) should be less than the expected noise criterion level of the particular area to be illuminated.

4.3.18 The lamps, fixtures and ballasts shall be as per latest edition of the relevant IEC or equivalent ANSI standards listed under Bibliography.

5.0 ILLUMINATION LEVELS

5.1 The yard (outdoor) lighting shall provide a minimum maintained (lifelong) illumination level of 20 lux on horizontal plane. Outdoor Transformer Bay (Room), Grounding transformer, Station service transformer, Oil immersed reactor, Capacitor bank, SVC and HVDC equipment shall have a minimum maintained (lifelong) illumination level of 100 lux.

5.2 The (indoor) building general fluorescent/LED lighting system shall provide the minimum maintained illumination level as per Table 25-2 below.

Table 25-2: Minimum Maintained Illumination Level

Location	Illumination Level (lux)
a. MV Switchgear Room and 69kV Indoor Live Tank Switchgear Cell	300
b. Relay and Control Room	400
c. SCADA/SAS Communication Room	300
d. Battery Room and HVAC Mechanical Room	200
e. Office/Workshop	500
f. Toilet	100
g. Basement/Cellars	100
h. Corridors	100
i. GIS Building	300
j. AC/DC Distribution Room	300
k. Gate House	300

l. Stairs	100
m. Trenches (more than1m deep)	100
n. Fire pump room (if any)	300
o. Fire Protection Equipment Room	200

5.3 The minimum maintained lux levels for areas not specified in clause 5.2 above shall be as specified in Table-111-1 (Illumination levels) of National Electrical Safety code (NESC) or IES Lighting Handbook.

5.4 The emergency lighting system as per clause 4.2.2 & 4.2.3 shall be provided average maintained illumination level of 10 lux measured at the floor.

5.5 Building entrance outside lighting shall have an illumination level of 50 lux.

6.0 POWER RECEPTACLES

6.1 Outdoor Power Receptacles

6.1.1 One (1), 3phase, 400 V_{ac}, 200 A, outdoor power receptacle complete with male/female connector assembly with a fuse-disconnect switch/MCCB housed in weather proof IP54 enclosure, shall be provided at each Power transformer, Grounding transformer, Station service transformer, Oil immersed reactor bay for power supply to oil treatment/filtration equipment, at an approved and convenient location. The receptacles shall be fed from the main LV AC distribution panel or the LV auto-transfer switch panel. Receptacle shall have a retained screw cap cover and an integral ON/OFF switch. Plug shall be equipped with captive screwed locking rings for retention in the socket. It shall not be possible to remove the plug with the socket switched ON. Receptacles shall be grounded per National Electrical Code (NFPA 70) Article 250.

6.1.2 One (1), 230 V_{ac}, 20 A, galvanized finish IP54 (per IEC 60529), weather proof steel clad receptacle/outlets with spring flap cover shall be provided on each lighting pole and at each Circuit breaker, Capacitor bank, SVC, HVDC, HVAC equipment, Transformer area, Grounding transformer, Station service transformer and Oil immersed reactor bay for supplying power to inspection lamps. Receptacles shall be installed strategically, for the attachment of portable cords of maximum length 30 meters to reach transformers and circuit breakers. The 230 V_{ac} receptacle outlets shall have tandem slots with grounding. All receptacle outlets shall be UL listed, or approved type and suitable for use with 3-pin fused plugs. All receptacles shall be fed from AC lighting panels.

6.1.3 Minimum one number of 400 V_{ac}, 63 A, 3 phase weather proof IP54 (per IEC 60529) welding receptacles shall be provided at approved strategic locations and within 30 meters from HV circuit breaker, Power transformers, Grounding transformer, Station service transformer, Oil immersed reactor, Capacitor bank, HVAC equipment, SVC/HVDC equipment and shall be mounted on structure/bracket or equipment supporting columns. The

receptacles shall generally conform to the other applicable requirements of clause 6.1.1 and shall be fed from the main LV AC distribution panel.

6.1.4 Receptacle outlets and attachment plugs for different voltages shall not be interchangeable. All power outlets shall be provided with stainless steel nameplate.

6.2 Building (Indoor) Receptacle Outlets

6.2.1 The 230 V_{ac}, 16/20 A receptacle outlets as applicable shall have tandem slots with grounding as per applicable IEC/NEC. All receptacle outlets shall be UL listed or approved type and suitable for use with 3-pin fused plugs. All receptacles shall be fed from AC lighting panels.

6.2.2 Receptacle outlets for different voltages shall not be interchangeable. Receptacle outlets shall be flush mounted type where wiring installations are concealed. Receptacle front cover plate provided in office, control and relay room shall be insulated type. Receptacle front cover plate shall be of stainless steel in the workshop.

6.2.3 Height of the receptacles above finished floor level shall be 1 (one) meter in workshop and 300 mm in other areas.

6.2.4 Location of receptacles shall be coordinated with furniture and equipment arrangement.

6.2.5 All receptacles in the basement, transformer bay (room), fire pump room (if any) and toilets shall be have (5mA sensitivity) Ground Fault Circuit Interrupter (GFCI) protection. Receptacles used for portable lighting and small power shall also have GFCI protection.

6.2.6 One (1), 3phase, 400 V_{ac}, 40 A power receptacle each complete with male/female connector assembly with a fuse-disconnect switch/MCCB shall be provided in GIS room per 4 bays of GIS for the use of gas handling equipment. The receptacle shall be fed from the main LV AC distribution panel. Receptacle shall have a retained screw cap cover and an integral ON/OFF switch. Plug shall be equipped with captive screwed locking rings for retention in the socket. It shall not be possible to remove the plug with the socket switched ON. Receptacle shall be grounded per NEC Article 250.

7.0 LIGHTING PANELS

7.1 Lighting and power receptacle panels and DC emergency lighting panel shall be in compliance to 31-TMSS-02 and TES-P-119.30.

7.2 Local distribution boards shall be installed in an approved location as required and in all individual buildings from which final sub-circuits shall supply the lighting circuits and all other current consuming accessories, receptacles etc.

7.3 The lighting panel shall be painted as per 01-TMSS-01.

8.0 WIRES, CABLES, CONDUITS AND JUNCTION BOXES

8.1 Conductor size for power receptacles and lighting circuits shall be selected taking into consideration the voltage drop, but shall not be smaller than $4mm^2$ stranded copper. For LED lamp circuits $2.5mm^2$ stranded copper shall be considered with prior approval of National Grid Saudi Arabia.

8.2 Wires shall be PVC insulated , 90°C maximum operating temperature, voltage up to 450V/750V, stranded annealed copper conductor, conforming to IEC 60227-1 and 60227-3. Thermoplastic High Heat-resistant Nylon Coated (THHN) cable shall be used for indoor/outdoor with prior approval of National Grid Saudi Arabia.

8.3 The cables for lighting and small power installations shall be per 11-TMSS-10. The cable sizing and installation shall be per TES-P-119.29 and TES-P-104-05.

8.4 All cabling and wiring shall be installed in conduits. There shall be no cable joints in any cable circuit. The size of the conduit shall be selected per Table 1, chapter 9 of NFPA 70, based on this following are percentage conduit fill:

53% for 1 conductor/cable
31% for 2 conductors/cables
40% for 3 or more conductors/cables

8.5 Concealed wiring shall be provided unless otherwise specified and ASTM D 1785, Schedule 40 Rigid PVC conduits of minimum 25mm size (inside diameter) with all fittings securely fixed on the same shall be used.

8.6 PVC conduits of minimum 50mm size shall be used for underground applications. They shall be type “DB” for direct buried and type “EB” for concrete encased installation.

8.7 Rigid galvanized steel conduits of minimum 25 mm size shall be used in all exposed locations/places other than concealed wiring. All couplings between sections of conduit shall be screwed in such a way that a perfect electrical and mechanical continuity is obtained in the conduit. Maximum spacing of saddles shall be 1 (one) meter. Heavy duty saddles shall be used for fixing the conduits to walls or ceiling. For ease of installation and to avoid crossing of exposed conduits, pipe sleeves may be used in cellar/basement beams, if applicable.

8.8 Draw-in boxes shall be provided at intervals not exceeding 12 meters in straight through runs and inspection bends / in 90 degree bends and boxes shall be located so as to facilitate access.

8.9 Where conduits cross building expansion joints, flexible couplings shall be provided. Joints under the concrete shall be avoided as far as possible.

8.10 Connection boxes shall be used where tees and connections are used.

8.11 AC and DC supply cables shall not be run in the same conduit or galvanized C-channels.

8.12 Galvanized sheet steel junction boxes with adequate number of outlets/ways, suitable for the rigid galvanized steel conduit entry shall be provided in the circuits wherever required. For embedded PVC conduits, PVC/cast iron/steel junction boxes shall be used. Grounding screw shall be provided in each junction box.

8.13 Where connections to apparatus are required to be flexible they shall be made in waterproof flexible metallic conduit. The ends of the flexible conduit shall be secured with approved glands.

9.0 GROUNDING OF LIGHTING EQUIPMENT

Grounding of lighting fixtures, lamp holders, lamps, receptacles and metal poles supporting lighting fixtures shall be per Article 250 and 410 of NEC (NFPA70).

10.0 BIBLIOGRAPHY

1. IEC 60081, "Double-capped Fluorescent Lamps – Performance specification".

2. IEC 60227-1, “Polyvinyl chloride insulated cables of rated voltages up to and including 450/750V- Part 1: General requirements”.

3. IEC 60227-3, “Polyvinyl chloride insulated cables of rated voltages up to and including 450/750V- Part 3: Non-sheathed cables for fixed wiring”.

4. IEC 60309-1, “Socket-outlets and coupler for industrial purpose - Part 1: General requirements”.

5. IEC 60529, "Degree of Protection provided by Enclosure (IP Code)".

6. IEC 60598-1, "Luminaires, Part 1: General Requirements and Tests".

7. IEC 60598-2, "Luminaires, Part 2: Particular Requirements".

8. IEC 60662, "High Pressure Sodium Vapour Lamps-Performance specifications".

9. IEC 60838-2-2, “Particular requirements – Connectors for LED-modules”.

10. IEC 60884-1, “Plug and socket-outlets for household and similar purposes- Part – 1: General requirements”.

11. IEC 60906-1, “IEC system of plugs and sockets – outlets for household and similar purpose-Part-1: Plugs and socket-outlets 16A, 250Vac”.

12. IEC 60921, “Ballast for tubular fluorescent lamps- performance requirement”.

13. IEC 60923, "Auxiliaries for lamps - Ballasts for discharge lamps (excluding tubular fluorescent lamps) - Performance Requirements".

14. IEC 60947-2, "Low-Voltage Switchgear and Controlgear Part 2: Circuit-breaker".

15. IEC 61048, "Auxiliaries for lamps- Capacitors for use in tubular fluorescent and other discharge lamp circuits- general and safety requirement".

16. IEC 61049, "Capacitors for use in tubular fluorescent and other discharge lamp circuits-performance requirement".

17. IEC 61347-1, "Lamp controlgear, Part 1: General and safety requirements".

18. IEC 61347-2-8, "Lamp Controlgear-Particular requirement for ballast for fluorescent lamps".

19. IEC 61347-2-9, "Lamp Controlgear-Particular requirements for ballasts for discharge lamps (excluding fluorescent lamps)".

20. IEC 61347-2-13, "Particular requirements for d.c and a.c supplied electronic controlgear for LED modules.".

21. IEC 61439, "Low Voltage Switchgear and Control Gear Assemblies, Part-1."

22. IEC 61547, "Equipment for general lighting purpose-EMC immunity requirements".

23. IEC 62031, "LED modules for general lighting – Safety specifications."

24. IEC 62384, "DC or AC supplied electronic controlgear for LED modules-Performance requirements".

25. IEC 62560, "Self-ballasted LED lamps for general lighting services by voltage >50V-safety specifications"

26. IEC/PAS62612 "Self-ballasted LED –lamps for general lighting services-Performance requirements".

27. IEC/PAS 62717, "LED modules for general lighting- Performance requirements".

28. IEC/PAS 62722-2-1, "Luminaire performance Part 2-1: Particular requirements for LED luminaires".

29. IEC 62776, "Double-capped LED lamps designed to retrofit linear fluorescent lamps - Safety Specifications".

30. ANSI/IEEE C.2, "National Electrical Safety Code".

31. ANSI/UL 1029, "High-Intensity-Discharge Lamp Ballasts".

32. ANSI/UL 542, "Fluorescent Lamp starters".

33. ANSI/UL 844, "Luminaries for Use in Hazardous (Classified) Locations".

34. ANSI/UL 924, "Emergency Lighting and Power Equipment".

35. ANSI/UL 935, "Fluorescent Lamp Ballasts".

36. ASTM D 229, "Standard Test Methods for Rigid Sheet and Plate Materials Used for Electrical Insulation".

37. NFPA 70, "National Electrical Code", Articles 250 and 410".

38. SEC/04, "Security Lighting" – Security Directives for Industrial Facilities from Kingdom of Saudi Arabia, Ministry of Interior High Commission for Industrial Security (HCIS)".

39. BS EN 12464-1, "Light and lighting-Lighting of work places, Part 1: Indoor work places".

40. BS 546, "Two-pole and earthing – pin plugs, socket-outlets and socket-outlet adaptors for AC (50-60Hz) circuits up to 250V".

41. IES Lighting Handbook, Illumination Engineering Society.

42. John P. Frier and Mary E. Gazley Frier, "Industrial Lighting Systems".